Public: Balance Sheets

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1 Balance Sheet: Total Asset

Cash & **Cash Equivalents**
+Short-Term Investments
+Accounts & Other Receivables
+Inventories
+Other Current Assets
=**Total Current Assets** (**TCA**)
+Property, Plant & Equipment (PP&E, Net)
+Goodwill
+Intangible Assets (Net)
+Other Non-Current Assets
=**Total Assets** (**TA**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

1.1 Total Assets

Across a sample universe of approximately 300 corporate entities spanning a 10-year period (9,600+ quarterly balance sheet records), reported top-level metrics exhibit near-perfect baseline integrity. This serves as good common scaler in both mathematical properties and econometrical properties Cooper et al. [2008].

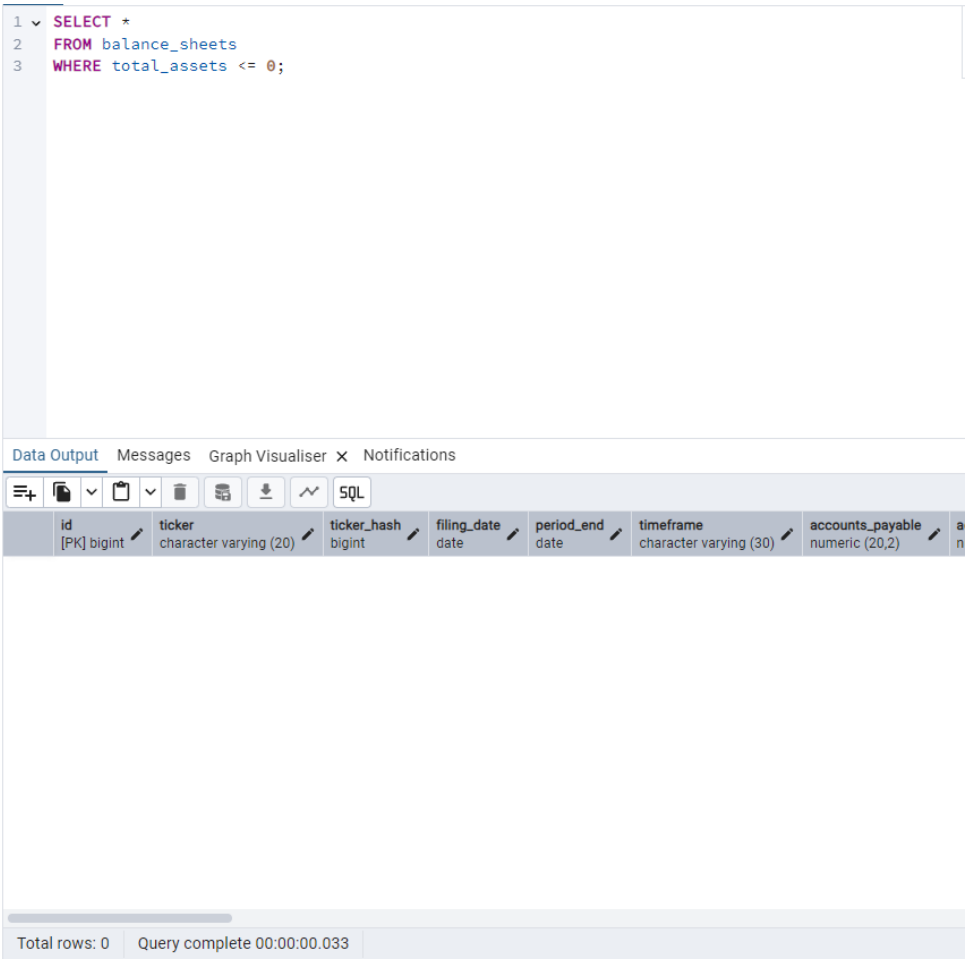


Figure 1: This validation returns exactly zero records, confirming that reported total_assets is reliably positive across all non-null filings.

1.2 Deduplication on Latest Filing

Multiple balance sheet observations frequently exist for a single (ticker, period_end) pair due to subsequent restatements, 10-K/A amendments, and preliminary earnings releases. Deduplicating by canonical ingestion order requires a partitioned window ranking by descending filing_date.

Listing 1 demonstrates the reconciliation check between annual and quarterly filings at identical period-ends.

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
)
SELECT
    a.ticker,
    a.period_end,
    a.total_assets AS annual_assets,
    q.total_assets AS quarterly_assets,
    (a.total_assets - q.total_assets) AS diff
FROM latest_balance_sheets a
INNER JOIN latest_balance_sheets q
    ON  a.ticker = q.ticker
    AND a.period_end = q.period_end
    AND q.timeframe = 'quarterly'
    AND q.rn = 1
WHERE a.timeframe = 'annual'
  AND a.rn = 1
ORDER BY diff DESC;
```

Listing 1: Annual vs. Quarterly Balance Sheet Reconciliation via Window Deduplication

Query | Query History

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
)
SELECT
    a.ticker,
    a.period_end,
    a.total_assets AS annual_assets,
    q.total_assets AS quarterly_assets,
    (a.total_assets - q.total_assets) AS diff
FROM latest_balance_sheets a
INNER JOIN latest_balance_sheets q
    ON  a.ticker = q.ticker
    AND a.period_end = q.period_end
    AND q.timeframe = 'quarterly'
    AND q.rn = 1
WHERE a.timeframe = 'annual'
  AND a.rn = 1
ORDER BY diff DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

	ticker character varying (20)	period_end date	annual_assets numeric (20,2)	quarterly_assets numeric (20,2)	diff numeric
1	A	2016-10-31	7794000000.00	7794000000.00	0.00
2	A	2017-10-31	8426000000.00	8426000000.00	0.00
3	A	2018-10-31	8541000000.00	8541000000.00	0.00
4	A	2019-10-31	9452000000.00	9452000000.00	0.00
5	A	2020-10-31	9627000000.00	9627000000.00	0.00
6	A	2021-10-31	10705000000.00	10705000000.00	0.00
7	A	2022-10-31	10532000000.00	10532000000.00	0.00
8	A	2023-10-31	10763000000.00	10763000000.00	0.00

Figure 2: Reconciliation with latest-filing deduplication (diff = 0).

```sql
SELECT
    a.ticker,
    a.period_end,
    a.total_assets AS annual_assets,
    q.total_assets AS quarterly_assets,
    (a.total_assets - q.total_assets) AS diff
FROM balance_sheets a
INNER JOIN balance_sheets q
    ON  a.ticker = q.ticker
    AND a.period_end = q.period_end
    AND q.timeframe = 'quarterly'
WHERE a.timeframe = 'annual'
ORDER BY diff DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

	ticker character varying (20)	period_end date	annual_assets numeric (20,2)	quarterly_assets numeric (20,2)	diff numeric
1	EP	2025-12-31	72748000000.00	65873000.00	72682127000.00
2	EP	2024-12-31	71407000000.00	123868000.00	71283132000.00
3	EP	2023-12-31	71020000000.00	92615616.00	70927384384.00
4	EP	2022-12-31	70078000000.00	71545075.00	70006454925.00
5	KO	2024-12-31	100549000000.00	99392000000.00	1157000000.00
6	TXN	2021-12-31	24676000000.00	24676000000.00	0.00
7	TXN	2022-12-31	27207000000.00	27207000000.00	0.00
8	TXN	2023-12-31	32348000000.00	32348000000.00	0.00
9	TXN	2024-12-31	35509000000.00	35509000000.00	0.00

Total rows: 2444 Query complete 00:00:00.051

Figure 3: Severe variance induced by non-canonical or arbitrary record ordering.

Figure 4: Impact of filing-date deduplication policies on balance sheet reconciliation.

Query Query History

```sql
WITH latest_annuals AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
    WHERE timeframe = 'annual'
)
SELECT
    a.ticker,
    a.period_end,
    a.filing_date,
    a.total_assets
FROM latest_annuals a
WHERE a.rn = 1
  AND NOT EXISTS (
      SELECT 1
      FROM balance_sheets q
      WHERE q.ticker = a.ticker
        AND q.period_end = a.period_end
        AND q.timeframe = 'quarterly'
  )
ORDER BY a.ticker ASC, a.period_end DESC;
```

Data Output Messages Graph Visualiser Notifications

ticker character varying (20)	period_end date	filing_date date	total_assets numeric (20,2)

Total rows: 0 Query complete 00:00:00.218

Figure 5: Verification showing complete quarterly representation for every annual filing period.

1.3 Current Asset Mapping Deficiencies

While reported aggregates (total_current_assets, total_assets) remain reliable audited anchors, reconstructing these aggregates from standard reporting line items surfaces parser classification flaws.

Out of 9,600 quarterly entries, only a single manufacturing line-item parsing omission was detected: Northrop Grumman (`NOC`, period ended September 30, 2023), where an unmapped XBRL cash tag caused an omitted $2.082B line item against the correctly reported total_current_assets total (Figure 6).

Query Query History

```
    bs.ticker,
    bs.period_end,
    bs.cash_and_equivalents,
    bs.short_term_investments,
    bs.receivables,
    bs.inventories,
    bs.other_current_assets,
    bs.total_current_assets as tca,
    (COALESCE(bs.cash_and_equivalents, 0)
    + COALESCE(bs.short_term_investments, 0)
    + COALESCE(bs.receivables, 0)
    + COALESCE(bs.inventories, 0)
    + COALESCE(bs.other_current_assets, 0)
    )  as r_tca,
    (COALESCE(bs.cash_and_equivalents, 0)
    + COALESCE(bs.short_term_investments, 0)
    + COALESCE(bs.receivables, 0)
    + COALESCE(bs.inventories, 0)
    + COALESCE(bs.other_current_assets, 0)
    ) - bs.total_current_assets AS diff



FROM latest_balance_sheets bs
WHERE bs.timeframe = 'quarterly'
  AND bs.rn = 1
    AND (
    (COALESCE(bs.cash_and_equivalents, 0)
    + COALESCE(bs.short_term_investments, 0)
    + COALESCE(bs.receivables, 0)
    + COALESCE(bs.inventories, 0)
    + COALESCE(bs.other_current_assets, 0)
    ) - bs.total_current_assets) <> 0
ORDER BY diff DESC;
```

Data Output Messages Graph Visualiser Notifications

Showing rows: 1 to 1 Page No: 1

	ticker character varying (20)	period_end date	cash_and_equivalents numeric (20,2)	short_term_investments numeric (20,2)	receivables numeric (20,2)	inventories numeric (20,2)	other_current_assets numeric (20,2)	tca numeric (20,2)	r_tca numeric	diff numeric
1	NOC	2023-09-30	[null]	0.00	8034000000.00	1355000000.00	1679000000.00	13150000000.00	11068000000.00	-2082000000.00

Total rows: 1 Query complete 00:00:00.046

Figure 6: Northrop Grumman (`NOC`) line-item ingestion omission in Q3 2023.

1.4 Total Asset Component Reconciliation

Listing 2 evaluates the discrepancy between reported total assets and reconstructed assets:

$$\text{Asset Coverage} = \frac{\text{Current Assets} + \text{Net PP\&E} + \text{Goodwill} + \text{Net Intangibles} + \text{Other Assets}}{\text{Reported Total Assets}} \tag{1}$$

```
1
2  WITH latest_balance_sheets AS (
3      SELECT
4          *,
5          ROW_NUMBER() OVER (
6              PARTITION BY ticker, period_end, timeframe
7              ORDER BY filing_date DESC
8          ) AS rn
9      FROM balance_sheets
10 ),
11 ratios as (
12     select
13         ticker,
14         period_end,
15         timeframe,
16         COALESCE(cash_and_equivalents / NULLIF(total_assets, 0), 0) as a_cash,
17         COALESCE(short_term_investments / NULLIF(total_assets, 0), 0) as a_short_invest,
18         COALESCE(receivables / NULLIF(total_assets, 0), 0) as a_receiv,
19         COALESCE(inventories / NULLIF(total_assets, 0), 0) as a_invent,
20         COALESCE(other_current_assets / NULLIF(total_assets, 0), 0) as a_other_cassets,
21         COALESCE(total_current_assets / NULLIF(total_assets, 0), 0) as
       a_total_current_assets,
22         COALESCE(property_plant_equipment_net / NULLIF(total_assets, 0), 0) as a_ppe_net
       ,
23         COALESCE(goodwill / NULLIF(total_assets, 0), 0) as a_goodwill,
24         COALESCE(intangible_assets_net / NULLIF(total_assets, 0), 0) as a_intangibles,
25         COALESCE(other_assets / NULLIF(total_assets, 0), 0) as a_other_assets
26     from latest_balance_sheets
27 )
28 SELECT
29     ticker,
30     period_end,
31     ROUND(a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles +
       a_other_assets, 4) AS r_total_assets_ratio,
32     1.0000 AS target_ratio,
33     ROUND((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles +
       a_other_assets) - 1.0, 4) AS ratio_discrepancy
34 FROM ratios
35   WHERE
36   ABS((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles + a_other_assets)
       - 1.0) > 0.03
37   AND ROUND((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles +
       a_other_assets), 4) <> 1.0000
38 ORDER BY ABS((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles +
       a_other_assets) - 1.0) DESC;
```

Listing 2: Total Asset Reconstruction Discrepancy Query

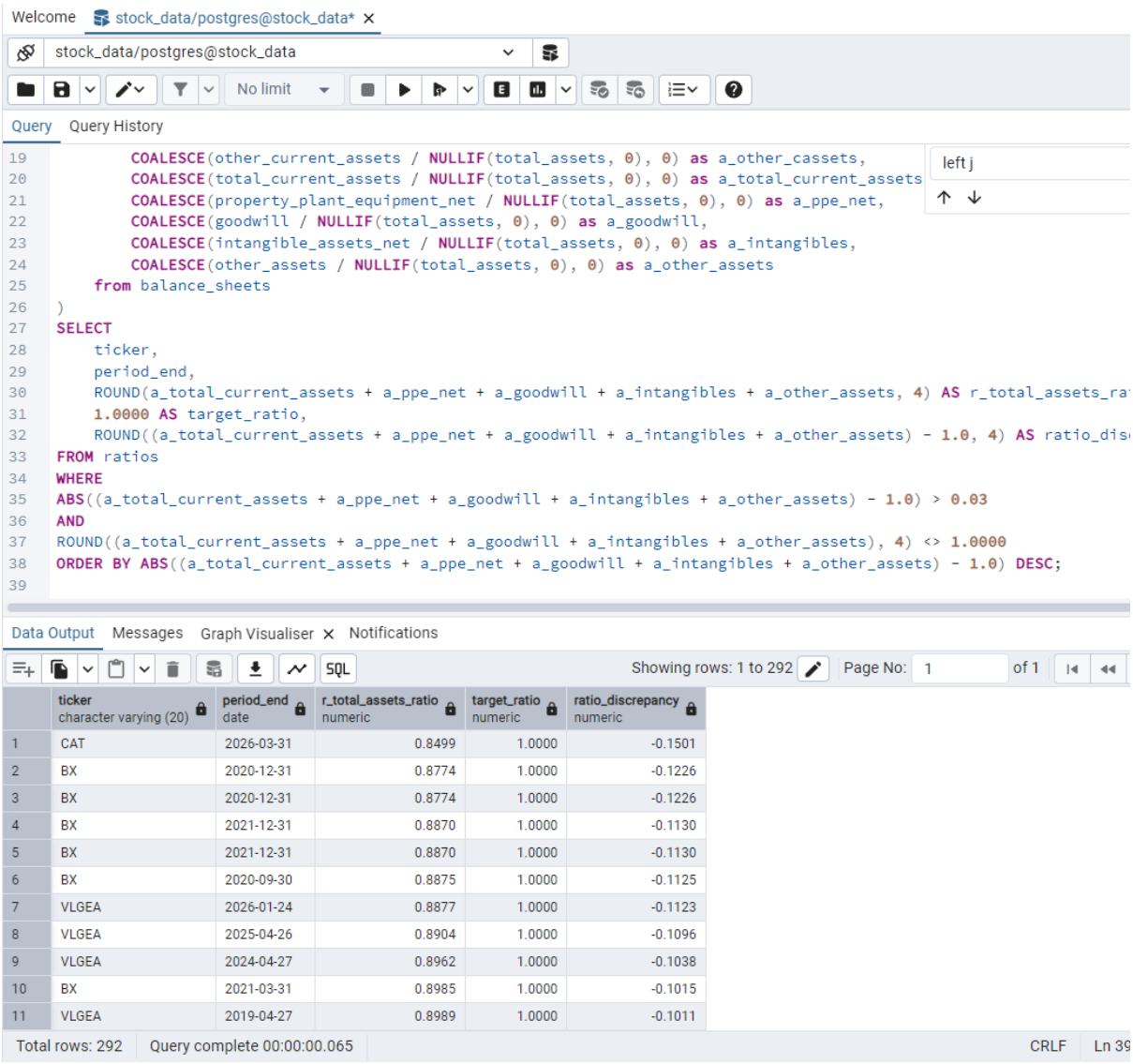


Figure 7: Distribution of 292 rows exhibiting reconstruction variance exceeding a 3% tolerance threshold, primarily driven by financial institutions, captive finance loans, and goodwill nesting.

2 Total Liabilities

Current Liabilities :
Current Debt (Short-Term Borrowings)
\+ Accounts Payable
\+ Accrued & Other Current Liabilities
=**Total Current Liabilities** (**TCL**)
+Long-Term Debt & Capital Lease Obligations
+Deferred Revenue (Non-Current / Other)
+Other Non-Current Liabilities
=**Total Liabilities** (**TL**)

Stockholders′ Equity :
Preferred Stock
\+ Common Stock
\+ Additional Paid-in Capital (APIC)
\+ Retained Earnings / (Deficit)
± Accumulated Other Comprehensive Income (AOCI)
− Treasury Stock
\+ Other Equity
=**Total Equity Attributable to Parent**
+Non-Controlling (Minority) Interest
=**Total Equity** (**TE**)

=**Total Liabilities** & **Equity** (**TL**&**E**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

2.1 Total Current Liabilities

Current Liabilities :
Current Debt (Short-Term Borrowings)
\+ Accounts Payable
\+ Accrued & Other Current Liabilities
=**Total Current Liabilities** (**TCL**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

Parallel to asset allocation, we deconstruct the right-hand side of the balance sheet into its primary funding sources: operating liabilities, debt financing, and shareholders' equity. Following standard corporate finance nomenclature [BARUCH-FIN3000, 2019]

2.2 Total Current Liabilities

Listing 3 details the audit query used to evaluate the internal consistency between reconstructed current liabilities and the reported aggregate.

```
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
```

```
        ) AS rn
    FROM balance_sheets
),
ratios as (
    select
        ticker,
        period_end,
        timeframe,
        COALESCE(debt_current , 0) as debt_current,
        COALESCE(accounts_payable , 0) as accounts_payable,
        COALESCE(accrued_and_other_current_liabilities , 0) as
    accrued_and_other_current_liabilities,
    COALESCE(deferred_revenue_current , 0) as deferred_revenue_current,
        COALESCE(total_current_liabilities , 0) as total_current_liabilities
    from latest_balance_sheets
  where timeframe = 'quarterly'
),
prep AS (
    SELECT
        ticker,
        period_end,
        total_current_liabilities,
        (
            COALESCE(debt_current, 0)
            + COALESCE(accounts_payable, 0)
            + COALESCE(accrued_and_other_current_liabilities, 0)
            + COALESCE(deferred_revenue_current, 0)
        ) AS r_tcl
    FROM ratios
    WHERE total_current_liabilities IS NOT NULL
      AND total_current_liabilities <> 0
),
evaluated AS (
    SELECT
        ticker,
        period_end,
        ROUND(r_tcl, 4) AS r_total_current_liabilities,
        ROUND(total_current_liabilities, 4) AS total_current_liabilities,
        1.0000 AS target_ratio,
        ROUND((r_tcl / total_current_liabilities) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_tcl / total_current_liabilities) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_current_liabilities,
    total_current_liabilities,
    target_ratio,
    ratio_discrepancy
FROM evaluated
WHERE abs_discrepancy >=0.01
ORDER BY abs_discrepancy DESC;
```

Listing 3: Total Current Liability Reconstruction Discrepancy Query

Query | Query History

```
40          ticker,
41          period_end,
42          ROUND(r_tcl, 4) AS r_total_current_liabilities,
43          ROUND(total_current_liabilities, 4) AS total_current_liabilities,
44          1.0000 AS target_ratio,
45          ROUND((r_tcl / total_current_liabilities) - 1.0, 4) AS ratio_discrepancy,
46          ABS((r_tcl / total_current_liabilities) - 1.0) AS abs_discrepancy
47      FROM prep
48  )
49  SELECT
50      ticker,
51      period_end,
52      r_total_current_liabilities,
53      total_current_liabilities,
54      target_ratio,
55      ratio_discrepancy
56  FROM evaluated
57  WHERE abs_discrepancy >=0.01
58  ORDER BY abs_discrepancy DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

	ticker character varying (20)	period_end date	r_total_current_liabilities numeric	total_current_liabilities numeric	target_ratio numeric	ratio_discrepancy numeric
1	VFF	2020-12-31	60209000.0000	72280000.0000	1.0000	-0.1670
2	PHM	2025-03-31	2521073000.0000	2983051000.0000	1.0000	-0.1549
3	PHM	2021-06-30	4655406000.0000	4770925000.0000	1.0000	-0.0242
4	KR	2017-01-28	12609000000.0000	12860000000.0000	1.0000	-0.0195
5	PBF	2016-03-31	1593032000.0000	1623003000.0000	1.0000	-0.0185
6	KR	2016-01-30	12750000000.0000	12971000000.0000	1.0000	-0.0170
7	KR	2016-05-21	12924000000.0000	13145000000.0000	1.0000	-0.0168
8	KR	2016-08-13	13300000000.0000	13521000000.0000	1.0000	-0.0163
9	KR	2016-11-05	13903000000.0000	14124000000.0000	1.0000	-0.0156
10	PBF	2016-09-30	2021301000.0000	2047733000.0000	1.0000	-0.0129
11	PBF	2016-06-30	1876988000.0000	1901518000.0000	1.0000	-0.0129
12	FSV	2016-12-31	173180000.0000	175122000.0000	1.0000	-0.0111

Total rows: 12 | Query complete 00:00:00.046

Figure 8: Current liability reconstruction discrepancies ($\geq 1\%$), affecting 12 of 9,439 quarterly observations.

Discrepancy Analysis

Across 9,439 quarterly filings evaluated, only 12 observations exhibited a component reconstruction discrepancy exceeding 1.0%. In each case, reported liabilities exceeded the reconstructed sum, indicating that omissions stem from industry-specific reporting lines rather than phantom liabilities:

- **PulteGroup (`PHM`) — Captive Finance & Unclassified Operations:** On 2025-03-31, reconstructed liabilities totaled \$2,521.07M versus reported \$2,983.05M (a gap of \$461.98M, or −15.5%). As a homebuilder operating a mortgage banking subsidiary (Pulte Mortgage), liabilities include warehouse credit facilities, customer escrow deposits, and obligations on consolidated inventory not owned. Because these captive finance obligations do not map to the primary four current liability columns, they fall out of the standard corporate sum.
- **Village Farms (`VFF`) — Current Operating Lease Liabilities (ASC 842):** On 2020-12-31, reconstructed liabilities totaled \$60.21M versus reported \$72.28M (a difference of \$12.07M, or −16.7%). Under ASC 842/IFRS 16, short-term lease obligations are categorized as `operating_lease_liability_current`. Because this column is absent from the standard four-column schema, capitalized lease payments create a structural deficit in agricultural and greenhouse operators with heavy leased infrastructure.
- **Kroger (`KR`) — Unmapped Accruals & Taxes Payable:** On 2017-01-28, reconstructed liabilities totaled \$12,609.00M versus reported \$12,860.00M (−1.9%). Retail grocery chains carry self-insurance reserves and income taxes payable due within one year. When vendor parsers do not bundle tax liabilities into `accrued_and_other_current_liabilities`, a mild deficit occurs.

2.3 Total Liabilities

Current Liabilities :
Current Debt (Short-Term Borrowings)
\+ Accounts Payable
\+ Accrued & Other Current Liabilities
=**Total Current Liabilities** (**TCL**)
+Long-Term Debt & Capital Lease Obligations
+Deferred Revenue (Non-Current / Other)
+Other Non-Current Liabilities
=**Total Liabilities** (**TL**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

Listing 4 details the audit query used to evaluate the internal consistency of reconstructed liabilities against reported total liabilities across quarterly balance sheets.

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
),
prep AS (
    SELECT
        ticker,
        period_end,
        total_liabilities,
        (
            COALESCE(total_current_liabilities, 0)
            + COALESCE(long_term_debt_and_capital_lease_obligations, 0)
            + COALESCE(other_noncurrent_liabilities, 0)
        ) AS r_tl
    FROM latest_balance_sheets
    WHERE
  timeframe = 'quarterly'
  AND rn = 1
    AND total_liabilities <> 0
    AND total_liabilities is NOT NULL
),
evaluated AS (
    SELECT
        ticker,
        period_end,
        ROUND(r_tl, 4) AS r_total_liabilities,
        ROUND(total_liabilities, 4) AS total_liabilities,
        1.0000 AS target_ratio,
        ROUND((r_tl / total_liabilities) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_tl / total_liabilities) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_liabilities,
    total_liabilities,
    target_ratio,
    ratio_discrepancy
FROM evaluated
WHERE abs_discrepancy >=0.01
ORDER BY abs_discrepancy DESC;
```

Listing 4: Total Liability Reconstruction Discrepancy Query

Query | Query History

```
        ticker,
        period_end,
        ROUND(r_tl, 4) AS r_total_liabilities,
        ROUND(total_liabilities, 4) AS total_liabilities,
        1.0000 AS target_ratio,
        ROUND((r_tl / total_liabilities) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_tl / total_liabilities) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_liabilities,
    total_liabilities,
    target_ratio,
    ratio_discrepancy
FROM evaluated
WHERE abs_discrepancy >=0.01
ORDER BY abs_discrepancy DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

	ticker character varying (20)	period_end date	r_total_liabilities numeric	total_liabilities numeric	target_ratio numeric	ratio_discrepancy numeric
1	CBNA	2026-06-30	7438000.0000	2008907000.0000	1.0000	-0.9963
2	CBNA	2025-03-31	6963000.0000	1575355000.0000	1.0000	-0.9956
3	CBNA	2025-06-30	6264000.0000	1288179000.0000	1.0000	-0.9951
4	CBNA	2025-09-30	6719000.0000	1371259000.0000	1.0000	-0.9951
5	CBNA	2025-12-31	7900000.0000	1581180000.0000	1.0000	-0.9950
6	CBNA	2024-12-31	6943000.0000	1256878000.0000	1.0000	-0.9945
7	CBNA	2026-03-31	13065000.0000	1743799000.0000	1.0000	-0.9925
8	COSO	2026-03-31	28480000.0000	2085624000.0000	1.0000	-0.9863
9	CCB	2022-03-31	49360000.0000	2625830000.0000	1.0000	-0.9812
10	CCB	2022-06-30	54756000.0000	2752061000.0000	1.0000	-0.9801
11	FGBI	2021-12-31	57739000.0000	2654231000.0000	1.0000	-0.9782
12	MVBF	2026-06-30	69934000.0000	3201310000.0000	1.0000	-0.9782
13	CCB	2026-03-31	118903000.0000	5160067000.0000	1.0000	-0.9770

Total rows: 940 | Query complete 00:00:00.065

Figure 9: Total liability reconstruction discrepancies ($\geq 1\%$), affecting 940 of 9,617 quarterly observations ($\approx 9.77\%$).

Analysis: Financial Institutions and Unclassified Balance Sheets

The query flagged 940 discrepancies across 9,617 quarterly observations ($\approx 9.77\%$). The most extreme deviations cluster between -98% and -99%, dominated by regional banks and financial holding companies:

- `CBNA` — Chain Bridge Bancorp (Commercial Bank)
- `COSO` — Coastal Financial / Community Banking Entity
- `MVBF` — MVB Financial Corp (Commercial Banking)
- `FGBI` — First Guaranty Bancshares (Regional Bank)

The standard corporate reconstruction formula assumes:

$$\text{Reconstr. Liabil.} = \text{Total Current Liabil.} + \text{Long-Term Debt} + \text{Other Noncurrent Liabil.} \quad (2)$$

This formulation systematically fails for commercial banks due to GAAP accounting standards (ASC 942):

1. **Omission of Customer Deposits (85%–95% of Liabilities):** A bank's core liabilities consist of customer deposits (checking, savings, money market accounts, CDs), repo facilities, and Federal Home Loan Bank (FHLB) advances. These instruments are categorized neither as corporate current liabilities nor as funded long-term debt. Because the schema lacks a dedicated `deposits` column, the vast majority of the liability structure is entirely omitted.

2. **Absence of Classified Balance Sheets:** Financial institutions operate with unclassified balance sheets that do not segregate assets or obligations into "current" versus "noncurrent" maturity buckets. Consequently, `total_current_liabilities` is typically null or zero, leaving only incidental holding-company obligations (such as minor corporate notes) captured by the formula.

3 Total Equity

	Stockholders′ Equity :
	Preferred Stock
+	Common Stock
+	Additional Paid-in Capital (APIC)
+	Retained Earnings / (Deficit)
±	Accumulated Other Comprehensive Income (AOCI)
−	Treasury Stock
+	Other Equity
=	**Total Equity Attributable to Parent**
+	Non-Controlling (Minority) Interest
=	**Total Equity** (**TE**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
),
prep AS (
    SELECT
        ticker,
        period_end,
        total_equity,
        (
            COALESCE(preferred_stock, 0)
            + COALESCE(common_stock, 0)
            + COALESCE(additional_paid_in_capital, 0)
            + COALESCE(retained_earnings_deficit, 0)
            + COALESCE(accumulated_other_comprehensive_income, 0)
            + COALESCE(treasury_stock, 0)
            + COALESCE(other_equity, 0)
            + COALESCE(noncontrolling_interest, 0)
        ) AS r_te
    FROM latest_balance_sheets
    WHERE timeframe = 'quarterly'
      AND rn = 1
      AND total_equity <> 0
      AND total_equity IS NOT NULL
),
evaluated AS (
    SELECT
        ticker,
        period_end,
        ROUND(r_te, 4) AS r_total_equity,
        ROUND(total_equity, 4) AS total_equity,
        1.0000 AS target_ratio,
        ROUND((r_te / total_equity) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_te / total_equity) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_equity,
    total_equity,
    target_ratio,
    ratio_discrepancy
```

```
FROM evaluated
WHERE abs_discrepancy >=0.00
ORDER BY abs_discrepancy DESC;
```

Listing 5: Total Equity Reconstruction Discrepancy Query

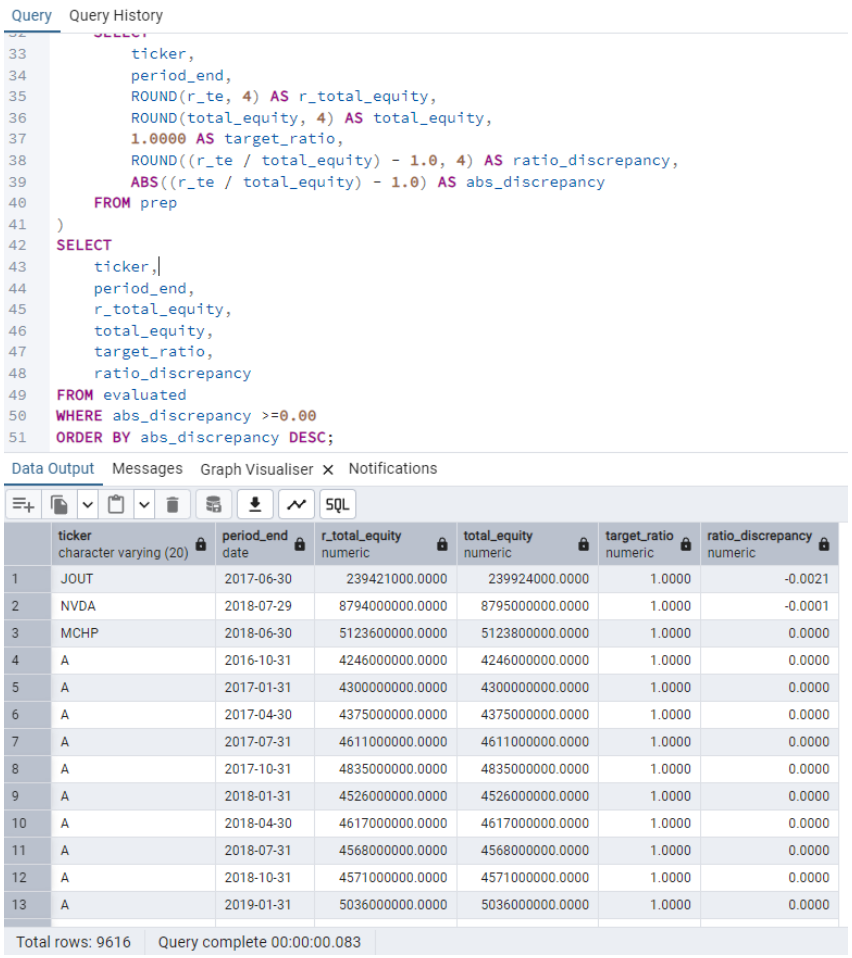


Figure 10: 0 rows affected, only 2 rounding error from 2017-2018

4 Total Total Asset, Liabilities, and Equity

Total Liabilities (**TL**)

+**Total Equity** (**TE**)

=**Total Liabilities** & **Equity** (**TL**&**E**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

Listing 6 details the verification query used to test whether r_ta = Total Assets within a 1.0% relative tolerance across canonical quarterly filings.

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
),
prep AS (
    SELECT
        ticker,
        period_end,
        total_assets,
        total_equity,
        total_liabilities,
        (
            COALESCE(total_equity, 0)
            + COALESCE(total_liabilities, 0)
        ) AS r_ta
    FROM latest_balance_sheets
    WHERE timeframe = 'quarterly'
      AND rn = 1
      AND total_assets IS NOT NULL
      AND total_assets <> 0
),
evaluated AS (
    SELECT
        ticker,
        period_end,
        ROUND(r_ta, 4) AS r_total_assets,
        ROUND(total_assets, 4) AS total_assets,
        1.0000 AS target_ratio,
        ROUND((r_ta / total_assets) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_ta / total_assets) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_assets,
    total_assets,
    target_ratio,
    ratio_discrepancy
FROM evaluated
-- WHERE abs_discrepancy > 0.01
ORDER BY abs_discrepancy DESC;
```

Listing 6: Balance Sheet Identity Reconciliation Query ($A = L + E$)

Discrepancy Analysis: Mezzanine and Redeemable Equity

Across 9,617 deduplicated quarterly records, 417 observations ($\approx 4.34\%$) failed the identity $(TL + TE)/TA = 1.0 \pm 0.01$. Rather than signaling mathematical non-compliance by the reporting entities, these deviations isolate an architectural limitation of two-sided schemas when processing securities subject to conditional redemption:

Query Query History

```
        ticker,
        period_end,
        ROUND(r_ta, 4) AS r_total_assets,
        ROUND(total_assets, 4) AS total_assets,
        1.0000 AS target_ratio,
        ROUND((r_ta / total_assets) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_ta / total_assets) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_assets,
    total_assets,
    target_ratio,
    ratio_discrepancy
FROM evaluated
WHERE abs_discrepancy > 0.01
ORDER BY abs_discrepancy DESC;
```

Data Output Messages Graph Visualiser Notifications

	ticker character varying (20)	period_end date	r_total_assets numeric	total_assets numeric	target_ratio numeric	ratio_discrepancy numeric
1	CNTX	2021-09-30	-18805319.0000	2259304.0000	1.0000	-9.3235
2	CVRX	2021-06-30	-274921000.0000	55062000.0000	1.0000	-5.9929
3	CCIV	2020-12-31	-1091395000.0000	1402681000.0000	1.0000	-1.7781
4	CCIV	2021-06-30	1966288.0000	2071966288.0000	1.0000	-0.9991
5	CCIV	2021-03-31	3181075.0000	2073181075.0000	1.0000	-0.9985
6	IRHO	2026-05-31	493985.0000	234030433.0000	1.0000	-0.9979
7	IRHO	2026-02-28	962821.0000	232424677.0000	1.0000	-0.9959
8	FCUV	2025-12-31	2939230.0000	8885514.0000	1.0000	-0.6692
9	STXS	2017-12-31	4706728.0000	10667203.0000	1.0000	-0.5588
10	MDXG	2023-03-31	75665000.0000	168159000.0000	1.0000	-0.5500
11	MDXG	2022-06-30	76173000.0000	168667000.0000	1.0000	-0.5484
12	MDXG	2022-12-31	78936000.0000	171430000.0000	1.0000	-0.5395
13	MDXG	2022-09-30	80304000.0000	172798000.0000	1.0000	-0.5353

Figure 11: Balance sheet identity discrepancies ($> 1\%$), affecting 417 of 9,617 quarterly observations ($\approx 4.34\%$).

- **Pre-Merger SPACs (`CCIV`, `IRHO`):** For Churchill Capital Corp IV (`CCIV`) on 2021-06-30, reported total assets equaled \$2,071.97M, whereas reconstructed assets totaled only \$1.97M (a ratio discrepancy of -0.9991, or -99.9%). Under SEC Rule 5-02.27 of Regulation S-X and ASC 480-10-S99, Class A shares subject to redemption must be classified outside of permanent stockholders' equity. Because SPAC trust proceeds reside in this "mezzanine" section, formulas summing only permanent liabilities and permanent equity omit $> 99\%$ of the capital structure.

- **Negative Reconstructed Assets in Early-Stage Biotechs (`CNTX`, `CVRX`):** Context Therapeutics (`CNTX`) reported assets of \$2.26M against a reconstructed total of $-\$18.81M$ (discrepancy of -9.3235). Similarly, CVRx (`CVRX`) showed \$55.06M in assets against $-\$274.92M$ in reconstructed assets. Prior to IPO, venture-backed biotechs issue convertible preferred stock with conditional redemption rights (categorized under mezzanine equity) to fund negative operating cash flows. The accumulated deficit in permanent retained earnings drives permanent stockholders' equity deeply negative. Omitting the positive mezzanine tranche leaves a negative equity balance that overwhelms liabilities, yielding a mathematically negative reconstructed asset total.

- **Post-DeSPAC Complex Capital Structures (`LCID`):** Lucid Group (`LCID`) on 2026-06-30 exhibited a -37.75% deficit (\$4,792.57M reconstructed vs. \$7,698.83M reported assets) driven by redeemable noncontrolling interests and convertible senior preferred shares issued post-business combination.

Systemic Resolution: To the Securities and Exchange Commission, these balance sheets balance to the penny under the tripartite standard:

$$\text{Total Assets} = \text{Total Liabilities} + \text{Temporary / Mezzanine Equity} + \text{Permanent Equity} \quad (3)$$

When third-party relational schemas flatten the right-hand side into only `total_liabilities` and permanent `total_equity`, mezzanine equity vanishes. Quantitative factor pipelines should either explicitly incorporate `temporary_equity` into Equation (**??**) or exclude SIC 6770 (Blank Check / SPAC entities) and pre-revenue clinical development issuers from standard industrial screens.

References

BARUCH-FIN3000. *Principles of Finance: FIN 3000 (Baruch College)*. McGraw-Hill Education, New York, NY, 2019. ISBN 9781260301182. Custom edition prepared for the Bert W. Wasserman Department of Economics and Finance, Zicklin School of Business, Baruch College.

Michael J. Cooper, Huseyin Gulen, and Michael J. Schill. Asset growth and the cross-section of stock returns. *The Journal of Finance*, 63(4):1609–1651, 2008. ISSN 00221082, 15406261. URL http://www.jstor.org/stable/25094485.